

(QAOA) -- both use a loop of iterations where classical optimisation techniques are used, and a quantum circuit to evaluate the potential solution.

Here's a simple form of a hybrid workflow:

- The classical system processes the input data.
- The quantum circuit is generated and run on a quantum processor.
- The quantum processor runs the circuit and sends back the measurements.
- The classical system optimises the parameters.
- The process is repeated until convergence is reached.

Such complex workflows are managed by complex orchestration tools like Kubernetes that help in managing distributed processes over CPUs, GPUs, and quantum computing. The other prominent player in the hybrid computing arena is high-performance computing, which is utilised for simulations, data processing, and creating intricate machine learning models.

The potential application areas for hybrid quantum and classical computing are:

- Drug discovery and molecular simulations
- Financial portfolio optimisation
- Logistics and supply chain optimisation
- Cryptography and cybersecurity research

With advancements in the hardware of quantum computers, hybrid infrastructures will play a significant role in bridging the gap between experimental quantum computers and practical application areas.

The future: Convergence of AI, HPC, and quantum computing in open cloud

The convergence of high-performance computing with AI and quantum technology represents one of the most interesting advances in computing

today. Each of these three areas is focused on different types of computational problems, and together, they create the opportunity to develop new types of applications. AI has the capability to assist in the optimisation of quantum algorithms, which can improve the design of quantum circuits and decrease the noise generated by quantum computing operations. On the other hand, quantum computing has the possibility of speeding up machine learning tasks such as optimisation and probabilistic modeling. The form of cloud infrastructure will provide the foundation for the unification of these three technologies into a single unified computing platform.

Open source ecosystems are providing the flexibility necessary to integrate a wide diversity of tools, frameworks and hardware architectures into a common unified computing platform. In the future, many companies may run a hybrid cloud in which workloads using AI will execute on GPU clusters, while simulations executed using HPC systems will run on specialised compute nodes and jobs submitted to remote quantum processors via cloud-based APIs. Developers will access all three types of resources through unified orchestration frameworks that

will hide the complexities of using these disparate technologies from the application developer.

Open communities have been instrumental in the development of open source software and cloud computing, along with container orchestration and distributed systems, thus far. As quantum and artificial intelligence technologies become more established as research areas, there is a widespread expectation that community development will continue to encourage innovation, transparency, and access through these types of developments. While scalable quantum computers are still years away from being developed as an enterprise computing platform, the groundwork for hybrid AI/quantum infrastructures is already being laid by individuals who are experimenting with these technologies. By getting involved early in the exploration of these technologies today, engineers, developers, and researchers will be well positioned to be part of the next computing era.

The transition from artificial intelligence cloud computing to quantum computing infrastructure has barely begun, but many of the leaders in this area are working with open source platforms to ensure that tomorrow's computing capabilities will be available to everyone. **END** 

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